

ROUTING LAB PRACTICAL

IP ADDED TO ROUTER INTERFACES AND ROUTE CHECK AT DIRECTLY CONNECTED ROUTE TABLE:-

labs.surennetwork.com/legacy/topology

Router

```
Router#sh ip int bri
Interface                                IP-Address      OK? Method Status      Prot
Ethernet0/0                             172.16.0.1      YES manual up          up
Ethernet0/1                             192.168.0.1     YES manual up          up
Ethernet0/2                             unassigned      YES TFTP  down        down
Ethernet0/3                             unassigned      YES TFTP  down        down

Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
       C       172.16.0.0/16 is directly connected, Ethernet0/0
       L       172.16.0.1/32 is directly connected, Ethernet0/0
       C       192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
       C       192.168.0.0/24 is directly connected, Ethernet0/1
       L       192.168.0.1/32 is directly connected, Ethernet0/1
Router#
```

50.0.0.1

Physical

01 : 00 : 00 START

31°C Sunny

Search

10:36 21-08-2026

IP ADDED TO PC'S ROUTE CHECK AT SYSTEM ROUTE TABLE:-

QEMU (Win1)

Select C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 172.16.0.10
Subnet Mask : 255.255.0.0
Default Gateway :

C:\Documents and Settings\user>route print

=====

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

Network Destination Netmask Gateway Interface Metric

127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1

172.16.0.0 255.255.0.0 172.16.0.10 172.16.0.10 10

172.16.0.10 255.255.255.255 172.0.0.1 172.0.0.1 10

172.16.255.255 255.255.255.255 172.16.0.10 172.16.0.10 10

224.0.0.0 240.0.0.0 172.16.0.10 172.16.0.10 10

255.255.255.255 255.255.255.255 172.16.0.10 172.16.0.10 1

=====

Persistent Routes:

None

C:\Documents and Settings\user>

QEMU (Win2)

Select C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 192.168.0.30
Subnet Mask : 255.255.255.0
Default Gateway :

C:\Documents and Settings\user>route print

=====

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

Network Destination Netmask Gateway Interface Metric

127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1

192.168.0.0 255.255.255.0 192.168.0.30 192.168.0.30 10

192.168.0.30 255.255.255.255 172.0.0.1 172.0.0.1 10

192.168.0.255 255.255.255.255 192.168.0.30 192.168.0.30 10

224.0.0.0 240.0.0.0 192.168.0.30 192.168.0.30 10

255.255.255.255 255.255.255.255 192.168.0.30 192.168.0.30 1

=====

Persistent Routes:

None

C:\Documents and Settings\user>

Physical

01 : 00 : 00

START

31°C Sunny

Search

10:34 21-08-2026

AFTER ROUTE CHECK PING THE DESTINATION IP GETTING A DISTINATION HOST UNREACHABLE ERROR DUE TO ROUTE DOES NOT MATCH:-

← → ↺ labs.surennetwork.com/legacy/topolog

☆ Paused ⋮

🔔 Notifications

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM...

Old Java ASD

PUTTY

TfRpd32

QEMU (Win1)

C:\WINDOWS\system32\cmd.exe

IP Address : 172.16.0.10
Subnet Mask : 255.255.0.0
Default Gateway :

C:\Documents and Settings\user>route print
=====

Interface List
0x1 MS TCP Loopback interface
0x2 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====

Active Routes:
Network Destination Netmask Gateway Interface Metric
127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1
172.16.0.0 255.255.0.0 172.16.0.10 172.16.0.10 10
172.16.0.10 255.255.255.255 127.0.0.1 127.0.0.1 10
172.16.255.255 255.255.255.255 172.16.0.10 172.16.0.10 10
224.0.0.0 240.0.0.0 172.16.0.10 172.16.0.10 10
255.255.255.255 255.255.255.255 172.16.0.10 172.16.0.10 1
=====

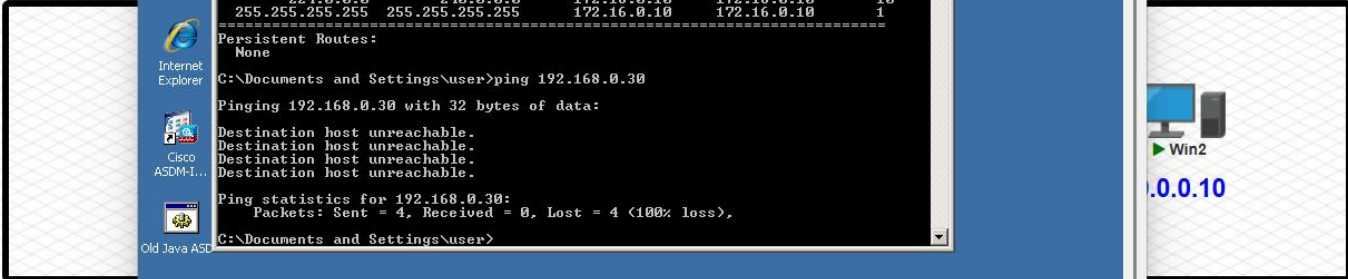
Persistent Routes:
None
C:\Documents and Settings\user>ping 192.168.0.30
Pinging 192.168.0.30 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 192.168.0.30:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\user>

Win2
0.0.10

Physical 🔍 🔍 ↺ 01 : 00 : 00 START

32°C Sunny Search 10:39 21-08-2026



MANUALLY ADD DESTINATION ROUTE AT PC1:-

← → ↺ labs.surennetwork.com/legacy/topolog

QEMU (Win1)

⌵ ⌵ ⌵

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASDM

PutTY

TRtpd32

Select C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:

Destination host unreachable.

Destination host unreachable.

Destination host unreachable.

Destination host unreachable.

Ping statistics for 192.168.0.30:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>route add 192.168.0.0 mask 255.255.255.0 172.16.0.1

C:\Documents and Settings\user>route print

=====

Interface List

0x1 MS TCP Loopback interface

0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

Table

Network Destination

Netmask

Gateway

Interface

Metric

1

10

10

10

1

10

1

Persistent Routes:

None

C:\Documents and Settings\user>

Win2

0.0.10

Physical

01 : 00 : 00

START

32°C

Sunny

Search

10:44

21-08-2026

AFTER MANNUALY ADD DESTINATION ROUTE PC 1 TO REACH DESTINATION IP GETTING REPLY FROM 192.168.0.30 REQUEST TIME OUT ERROR OCCURRED:-

PNETLab | Topology

labs.surennetwork.com/legacy/to

Ask Gemini

Paused

Notifications

QEMU (Win1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD...

PutTY

Tftpd32

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>route add 192.168.0.0 mask 255.255.255.0 172.16.0.1

C:\Documents and Settings\user>route print

=====

Interface List

0x1 MS TCP Loopback interface

0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

=====

Network Destination	Netmask	Gateway	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
172.16.0.0	255.255.0.0	172.16.0.10	172.16.0.10	10
172.16.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
172.16.255.255	255.255.255.255	172.16.0.10	172.16.0.10	10
192.168.0.0	255.255.255.0	172.16.0.1	172.16.0.10	1
224.0.0.0	240.0.0.0	172.16.0.10	172.16.0.10	10
255.255.255.255	255.255.255.255	172.16.0.10	172.16.0.10	1

=====

Persistent Routes:

None

C:\Documents and Settings\user>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.0.30:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>

Win2

60.0.0.10

Physical

01 : 00 : 00

START

32°C Sunny

Search

10:49 21-08-2026

MANUALLY ADD DESTINATION ROUTE AT PC2:-

← → ↺ labs.surennetwork.com/legacy/topology

☆ Paused

Notifications

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASD

PUTTY

Tftpd32

Win1

50.0.0.10

QEMU (Win2)

Select C:\WINDOWS\system32\cmd.exe

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1	
192.168.0.0	255.255.255.0	192.168.0.30	192.168.0.30	10	
192.168.0.30	255.255.255.255	127.0.0.1	127.0.0.1	10	
192.168.0.255	255.255.255.255	192.168.0.30	192.168.0.30	10	
224.0.0.0	240.0.0.0	192.168.0.30	192.168.0.30	10	
255.255.255.255	255.255.255.255	192.168.0.30	192.168.0.30	1	

Persistent Routes:
None

C:\Documents and Settings\user>route add 172.16.0.0 mask 255.255.0.0 192.168.0.1

C:\Documents and Settings\user>route print

Interface List

Interface	MS TCP Loopback interface
0x1	Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1	
172.16.0.0	255.255.0.0	192.168.0.1	192.168.0.30	10	
192.168.0.0	255.255.255.0	127.0.0.1	127.0.0.1	10	
192.168.0.30	255.255.255.255	192.168.0.30	192.168.0.30	10	
224.0.0.0	240.0.0.0	192.168.0.30	192.168.0.30	10	
255.255.255.255	255.255.255.255	192.168.0.30	192.168.0.30	1	

Persistent Routes:
None

C:\Documents and Settings\user>

Physical

01 : 00 : 00

START

32°C Sunny

Search

10:54 21-08-2026

MANNUALLY ROUTING TWO DIFFERENT NETWORK PC'S GETTING REACHABILITY OUTPUT

QEMU (Win1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD

PUTTY

Tftpd32

```
C:\WINDOWS\system32\cmd.exe

127.0.0.0      255.0.0.0      127.0.0.1      127.0.0.1      1
172.16.0.0     255.255.0.0    172.16.0.10    172.16.0.10    10
172.16.0.10    255.255.255.255 127.0.0.1      127.0.0.1      10
172.16.255.255 255.255.255.255 172.16.0.10    172.16.0.10    10
192.168.0.0    255.255.255.0   172.16.0.1      172.16.0.10    1
224.0.0.0      240.0.0.0      172.16.0.10    172.16.0.10    10
255.255.255.255 255.255.255.255 172.16.0.10    172.16.0.10    1

=====
Persistent Routes:
None

C:\Documents and Settings\User>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\User>
C:\Documents and Settings\User>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:

Reply from 192.168.0.30: bytes=32 time=2ms TTL=127
Reply from 192.168.0.30: bytes=32 time=1ms TTL=127
Reply from 192.168.0.30: bytes=32 time=1ms TTL=127
Reply from 192.168.0.30: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>
```

QEMU (Win2)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD

PUTTY

Tftpd32

```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>route print

=====
Interface List
=====
0x1 ..... MS TCP Loopback interface
0x2 ..50 86 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====

Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
127.0.0.0                  255.0.0.0        127.0.0.1         127.0.0.1         1
172.16.0.0                  255.255.0.0      192.168.0.1       192.168.0.30      1
192.168.0.0                 255.255.255.0    192.168.0.30      192.168.0.30      10
192.168.0.30               255.255.255.255   127.0.0.1         127.0.0.1         10
192.168.0.255              255.255.255.255   192.168.0.30      192.168.0.30      10
224.0.0.0                  240.0.0.0        192.168.0.30      192.168.0.30      10
255.255.255.255            255.255.255.255   192.168.0.30      192.168.0.30      1

=====
Persistent Routes:
None

C:\Documents and Settings\User>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=6ms TTL=127
Reply from 172.16.0.10: bytes=32 time=1ms TTL=127
Reply from 172.16.0.10: bytes=32 time=1ms TTL=127
Reply from 172.16.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

C:\Documents and Settings\User>
```

Physical

01 : 00 : 00

START

32°C Sunny

Search

10:56 21-08-2026

BOTH PC'S MANNUALY ADDED ROUTES DELETE:-

QEMU (Win1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD

PUTTY

Tftpd32

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>route delete 192.168.0.0

C:\Documents and Settings\user>route print

Interface List

=====

Active Routes:

=====

Network	Destination	Netmask	Gateway	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1	
172.16.0.0	255.255.0.0	172.16.0.10	172.16.0.10	10	
172.16.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10	
172.16.255.255	255.255.255.255	172.16.0.10	172.16.0.10	10	
224.0.0.0	240.0.0.0	172.16.0.10	172.16.0.10	10	
255.255.255.255	255.255.255.255	172.16.0.10	172.16.0.10	1	

Persistent Routes:

None

C:\Documents and Settings\user>

QEMU (Win2)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASD

PUTTY

Tftpd32

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>route delete 172.16.0.0

C:\Documents and Settings\user>route print

Interface List

=====

Active Routes:

=====

Network	Destination	Netmask	Gateway	Interface	Metric
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1	
192.168.0.0	255.255.255.0	192.168.0.30	192.168.0.30	10	
192.168.0.30	255.255.255.255	127.0.0.1	127.0.0.1	10	
192.168.0.255	255.255.255.255	192.168.0.30	192.168.0.30	10	
224.0.0.0	240.0.0.0	192.168.0.30	192.168.0.30	10	
255.255.255.255	255.255.255.255	192.168.0.30	192.168.0.30	1	

Persistent Routes:

None

C:\Documents and Settings\user>

Physical

01 : 00 : 00

START

32°C Sunny

Search

11:01 21-08-2026

REAL TIME SCENARIO MULTIPLE PC'S AVAILBLE TWO DIFFERENT NETWORK SO ADD A DEFAULT GATEWAY ADDED TO PC'S :-

The image displays two side-by-side screenshots of a Windows XP desktop environment, likely running in a QEMU virtual machine. Both windows show the 'Network Connections' window with the 'Local Area Connection Properties' dialog box open. The 'Internet Protocol (TCP/IP) Properties' dialog box is also open, showing the 'General' tab. In the left window (Win1), the IP address is set to 172.16.0.10, the subnet mask is 255.255.0.0, and the default gateway is 172.16.0.1. In the right window (Win2), the IP address is set to 192.168.0.30, the subnet mask is 255.255.255.0, and the default gateway is 192.168.0.1. Both windows show a 'Connected' status with a speed of 1.0 Gbps. The taskbar at the bottom of both windows includes the Start button, several application icons, and a system tray showing the time as 12:00 on 21-08-2026. A status bar at the very bottom of the image displays page information (Page: 9/9, Words: 105) and various utility icons like AI Spell Check, AI Co-Writing, AI Proofing, and Font Missing.

AFTER DEFAULT ROUTES CHECK BOTH PC'S IP IS REACHABILITY OUTPUT VIA DEFAULT GATEWAY:-

QEMU (Win1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASD

PUTTY

Tftpd32

Select C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>ncpa.cpl

=====

Interface List

0x1 MS TCP Loopback interface

0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.16.0.1	172.16.0.10	10
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
172.16.0.0	255.255.0.0	172.16.0.10	172.16.0.10	10
172.16.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
172.16.255.255	255.255.255.255	172.16.0.10	172.16.0.10	10
224.0.0.0	240.0.0.0	172.16.0.10	172.16.0.10	10
255.255.255.255	255.255.255.255	172.16.0.10	172.16.0.10	1

Default Gateway: 172.16.0.1

=====

Persistent Routes:

None

C:\Documents and Settings\User>ping 192.168.0.30

Pinging 192.168.0.30 with 32 bytes of data:

Reply from 192.168.0.30: bytes=32 time=2ms TTL=127

Reply from 192.168.0.30: bytes=32 time=1ms TTL=127

Reply from 192.168.0.30: bytes=32 time=1ms TTL=127

Reply from 192.168.0.30: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.0.30:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>

QEMU (Win2)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASD

PUTTY

Tftpd32

Select C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>route print

=====

Interface List

0x1 MS TCP Loopback interface

0x2 ...50 86 5f 09 50 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.0.1	192.168.0.30	10
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
192.168.0.0	255.255.255.0	192.168.0.30	192.168.0.30	10
192.168.0.30	255.255.255.255	127.0.0.1	127.0.0.1	10
192.168.0.255	255.255.255.255	192.168.0.30	192.168.0.30	10
224.0.0.0	240.0.0.0	192.168.0.30	192.168.0.30	10
255.255.255.255	255.255.255.255	192.168.0.30	192.168.0.30	1

Default Gateway: 192.168.0.1

=====

Persistent Routes:

None

C:\Documents and Settings\User>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=1ms TTL=127

Reply from 172.16.0.10: bytes=32 time=1ms TTL=127

Reply from 172.16.0.10: bytes=32 time=1ms TTL=127

Reply from 172.16.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\User>

Physical

01 : 00 : 00

START

33°C Sunny

Search

12:05 21-08-2026